BABU BANARASI DAS UNIVERSITY

Project Report on

Fake News Generator & Detector using Generative AI and NLP



Project Focus: Generative AI
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Acknowledgment

I would like to express my sincere gratitude to my esteemed mentor, **Mr. Harendra Singh Rajpoot,** for his unwavering guidance, insightful feedback, and constant support throughout the course of this project in **Generative AI**. His **mentorship** played a pivotal role in shaping this work into a valuable and enriching learning experience.

I am also deeply thankful to my peers for their continuous motivation, collaborative spirit, and **constructive suggestions**, which greatly contributed to the project's development.

A special note of appreciation goes to the creators and maintainers of **Google Gemini, Hugging Face**, and the broader open-source community. Their technological innovations and freely available tools provided the essential backbone that enabled the successful **execution of this project.** This endeavor would not have been possible without their dedication to knowledge sharing and open collaboration.

Abstract :

The rapid spread of misinformation on digital platforms poses a significant threat to society. This project demonstrates the dual capabilities of AI in both generating and detecting fake news. Using GPT-2, the system generates realistic fake news headlines and articles, highlighting the potential misuse of generative AI. Simultaneously, a BERT-based text classification model detects fake versus real news with high accuracy. This project emphasizes the ethical responsibility associated with generative AI and explores the intersection of natural language processing (NLP) and AI-driven content moderation.

2. Introduction

With the explosion of online media, misinformation has become a pressing global issue. Fake news can influence public opinion, manipulate political outcomes, and create social unrest. AI and NLP provide tools to both understand and mitigate this challenge:

- Generative AI (GPT-2): Can simulate human-like text, enabling the creation of fake news.
- Transformers-based classifiers (BERT): Effectively detect patterns and distinguish real news from fake content.

This project aims to create a system that demonstrates the potential of AI in both generating and detecting fake news, emphasizing the ethical considerations of AI deployment.

3. Methodology

3.1 Dataset and Pre-trained Models

- GPT-2: Pre-trained generative model for text generation.
- BERT: Pre-trained transformer model (bert-base-uncased) for binary classification (fake vs. real news).

3.2 Environment Setup

- Frameworks: PyTorch, Transformers library (Hugging Face), Gradio.
- Hardware: CPU/GPU (CUDA-enabled if available).

3.3 Text Generation with GPT-2

- Users provide a prompt, and GPT-2 generates a fake news article.
- Parameters for generation:
  - max_length=200 – maximum tokens generated.
  - no_repeat_ngram_size=2 – prevent repetition.
  - temperature=0.7, top_k=50, top_p=0.95 – control creativity and coherence.

3.4 Fake News Detection with BERT

- Input text is tokenized using BERT tokenizer.
- The classifier predicts whether the news is fake (0) or real (1).
- Softmax calculates confidence score.

3.5 User Interface

The frontend is built with:

- index.html – Input form for keywords/themes.
- results.html – Displays AI-generated captions, hashtags, and emojis.
- style.css – Defines layout, aesthetics, and responsive design.
- Jinja2 – Injects dynamic data into templates.

4. Implementation Details

The prototype implementation is contained in a Jupyter Notebook (e.g., fake_news_gen.ipynb).

4.1 Library Initialization

Libraries imported include:

- google.generativeai – for accessing Gemini models
- time – for retry logic in case of failed API calls
- textwrap – for formatting generated text

4.2 Retry & Error Handling

The system retries failed API calls up to 3 times with a 5-second interval.
Clear error messages are displayed if the API key is missing or the connection fails.

4.3 User Interface using Gradio

- Tabs for generating and detecting fake news.
- Textboxes for input, buttons for execution, output text display.

5. Results

- GPT-2 generates realistic fake news headlines and articles.
- BERT classifier detects fake vs. real news with confidence scores.

Example:

PromptGenerated Fake NewsDetection

"Scientists discover a new energy source"

"Scientists discover a new energy source that could revolutionize the world in 2025..."

˜ **Fake News (Confidence: 0.91)**

- Demonstrates AI's dual nature:
  - Creating misleading content
  - Combating misinformation

Figure: Screenshot of Gradio Interface showing generation and detection outputs

5. Conclusion

This project highlights the potential and ethical responsibility of generative AI. GPT-2 demonstrates how AI can generate realistic fake news, while BERT detects misinformation effectively. This dual-system provides a platform for research, awareness, and practical applications in combating fake news. Ethical deployment is crucial to ensure AI benefits society rather than causing harm.

Potential Future Enhancements:

- Larger Models: Upgrade GPT-2/BERT to GPT-3/4 or RoBERTa for better text generation and detection.
- Fine-tuning: Train models on news-specific datasets for improved realism and accuracy.
- Multi-class Detection: Classify news into fake, biased, satire, or real.

6. References

Hugging Face Transformers Documentation: https://huggingface.co/transformers

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